

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 03

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I Abinesh H (192424387) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

<i>Criteria</i>	<i>Conceptual Understanding</i>	<i>Accuracy of Answers</i>	<i>Application of Knowledge</i>	<i>Completeness of Response</i>	<i>Clarity & Presentation</i>	<i>Total</i>
<i>Weightage</i>	25%	25%	20%	15%	15%	
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Signature of the student: Abinesh H

Total Marks :

Annexure A

Short Answer Test

1. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

2. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using **KNN or Case-Based Learning**. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

3. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using **Locally Weighted Regression or RBF networks**. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Loan Default Detection Using Logistic Regression

Proposed Solution

A banking system can use **Logistic Regression** to predict whether a customer is likely to default on a loan.

Input features:

- Income
- Credit score
- Existing loan amount
- Debt-to-income ratio
- Number of previous defaults
- Transaction frequency
- Account balance
- Employment duration

How it works

Logistic Regression creates a **decision boundary** that separates customers into two classes:

- **0 → Non-default**
- **1 → Potential default**

The model calculates the probability of default. For example, if the predicted probability is greater than **0.5**, the customer can be classified as a potential defaulter.

Feature Selection

Important features can be selected using:

- Correlation analysis
- Feature importance
- Statistical significance
- Recursive feature elimination

Irrelevant or highly correlated features can be removed to improve model performance.

Adapting to New Data

As new customer repayment and transaction records become available:

1. New data is collected.
2. The data is cleaned and converted into the required features.
3. The model is periodically retrained.
4. Updated customer behavior is incorporated into the decision boundary.

Result: The system can continuously improve its ability to identify potential loan defaulters.

2. Intrusion Detection Using K-Nearest Neighbour

Proposed Solution

A cybersecurity system can use **K-Nearest Neighbour (KNN)** to classify network connections as **normal or suspicious**.

Network features:

- Connection duration
- Number of packets
- Bytes transferred
- Protocol type
- Source/destination port
- Failed connection attempts
- Connection frequency

Similarity Measurement

KNN compares a new network connection with previously observed network patterns.

A common similarity measure is **Euclidean distance**:

The system finds the **K most similar network connections** and uses their classes to classify the new connection.

Anomaly Detection

If most nearby patterns are known attacks, the new connection is classified as an **intrusion**.

For example:

New network pattern → Find nearest patterns → Vote → Normal/Attack

A pattern that is very different from known normal traffic can also be flagged for further investigation.

Updating with New Attack Patterns

When security analysts confirm a new attack:

1. The network pattern is stored.
2. It is labelled as a new attack type.
3. The case becomes part of the historical dataset.
4. Future connections can be compared with this new pattern.

Result: KNN can adapt naturally by adding newly labelled attack cases without completely redesigning the classification process.

3. Electricity Consumption Prediction Using Locally Weighted Regression

Proposed Solution

An energy management company can use **Locally Weighted Regression (LWR)** to predict electricity consumption.

Input features:

- Hour of day
- Day of week
- Temperature
- Humidity
- Previous consumption
- Appliance usage
- Holiday/weekend indicator

Local Learning

Unlike ordinary regression, LWR gives **higher weights to observations that are more similar to the current prediction point.**

For example, when predicting electricity usage at **7 PM on a hot weekday**, the model gives greater importance to historical observations from similar times and weather conditions.

A Gaussian weighting function can be used:

Here, τ controls how strongly the model focuses on nearby observations.

Handling Seasonal Patterns

The model can capture variations such as:

- Higher consumption during summer
- Lower consumption during mild weather
- Morning and evening peaks
- Weekend usage differences
- Holiday consumption changes

By learning locally, the model does not need one global relationship for all conditions.

Handling Noisy Data

Noisy observations can have less influence when they are far from the current prediction point.

Data preprocessing can further improve performance through:

- Outlier removal
- Missing-value treatment
- Feature scaling
- Moving averages

Result: LWR provides flexible predictions because it focuses on **local consumption patterns**, making it suitable for changing and seasonal electricity demand.